Growth-Driven or Redistributive? An Analysis of Chicago's TIF Spending

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Research Question

Chicago's Tax Increment Financing (TIF) program is designed to attract investment and support redevelopment in economically distressed neighbourhoods. When a TIF district is created, the property-tax level in that area is frozen at a baseline. As property tax rises, the additional tax revenue above that baseline is called the **tax increment**, and is retained inside the district to fund redevelopment.

In principle, this system should help channel public investment toward neighbourhoods with greater economic need. However, critics often argue that TIF districts benefit areas that are already experiencing growth. In that case, the program may reinforce existing inequalities rather than reduce them.

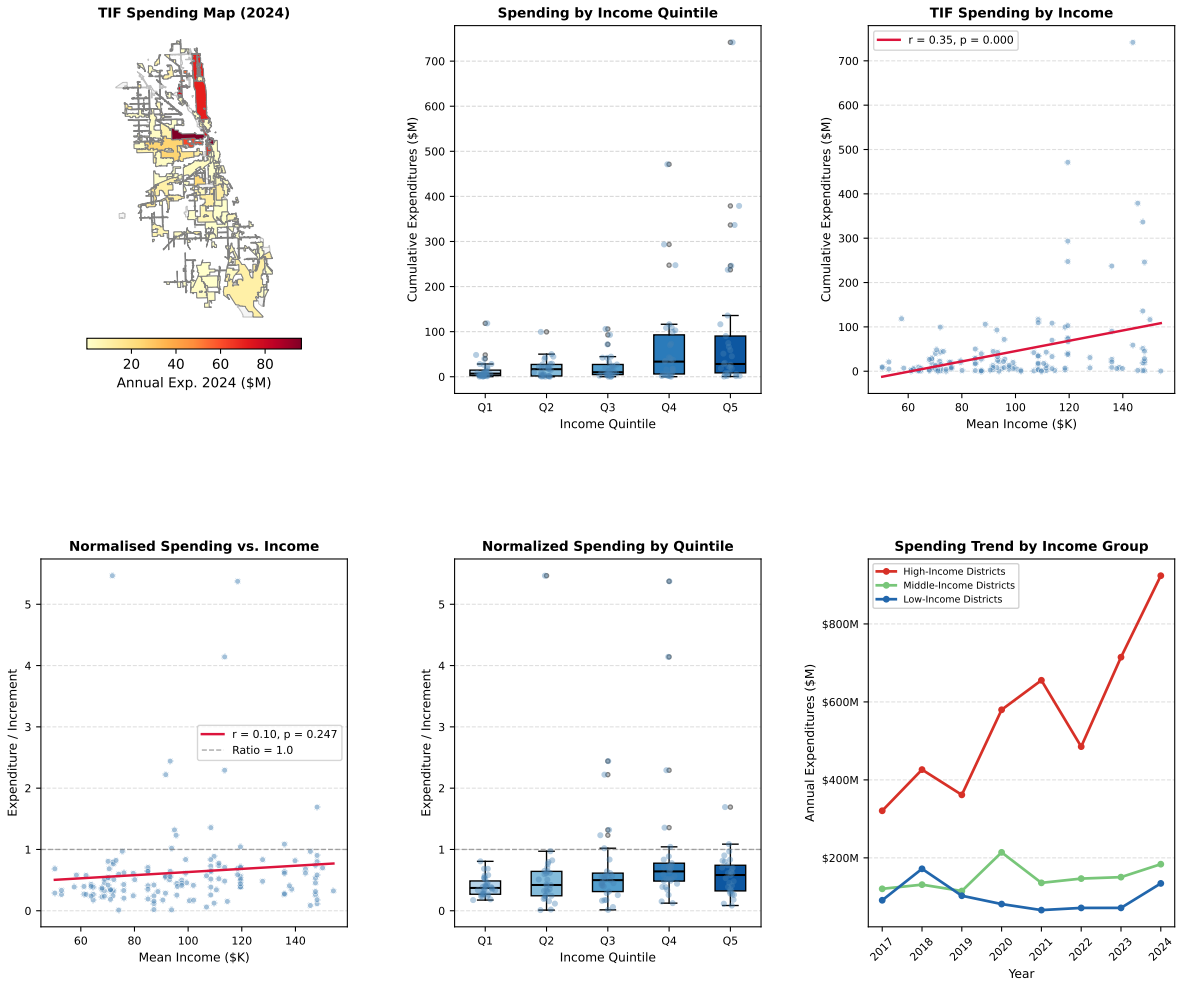
Therefore, this project asks: **is Chicago's TIF spending growth-driven or reditributive?**

Data & Approach

This analysis combines two main datasets, the Chicago TIF Annual Reports that provide yearly expenditure information for every TIF district in the city and the ACS 5-year community area data from the Chicago Data Portal that provides household income distributions for each of Chicago's 77 community areas. To map these districts, I merge the current TIF boundary file with a deprecated boundary dataset that includes districts repealed before March 2018.

To examine whether TIF spending is related to neighborhood income, I first compare total TIF expenditure across districts with the income levels of the communities where they are located. I then normalize the total TIF spending by the cumulative tax increment generated by each district to check whether the higher TIF spending is driven by higher tax revenue or by differences in how spending is distributed.

Static Visualisations



Row 1 — Left: TIF spending map (2024). Centre: cumulative expenditure by income quintile. Right: cumulative expenditure by income.

Row 2 — Left: normalized spending by income. Centre: normalized spending by quintile. Right: annual spending trend 2017–2024.

Absolute spending. High-income districts receive significantly more TIF funding in absolute terms. The map shows that spending is concentrated in the Loop and North Side. The boxplot shows a clear step-up from Q1 to Q5, and the scatter confirms a statistically significant positive relationship ($r = 0.35$, $p < 0.001$).

Normalised spending. Once spending is divided by tax increment, the relationship between income and spending becomes much weaker($r = 0.10$, $p = 0.247$). Wealthier districts spend

more in absolute terms only because they generate more tax increment. The trend chart shows the absolute gap has nonetheless grown from roughly 3× to 6× between 2017 and 2024.

Streamlit App

The interactive dashboard (`streamlit-app/app.py`) is designed for policymakers and city officials evaluating whether Chicago's TIF program is meeting its redistributive goals. The dashboard is deployed at https://final-project-group41.streamlit.app/ and contains five tabs:

- **District Map**: Switch between *Annual Expenditure*, *Community Income*, and *Normalised Spending* views, with a year slider.
- **Spending by Neighbourhood Income**: Boxplot of cumulative TIF expenditure across five income quintiles.
- **Does Income Predict Spending?**: Scatter plot with OLS trend line ($r = 0.35$, $p < 0.001$).
- **Is the Gap Fair? (Normalised View)**: Scatter and boxplot for the normalised ratio ($r = 0.10$, $p = 0.247$).
- **Is the Gap Growing?**: Line chart of annual expenditure by income tertile, 2017–2024.

Conclusion

Taken together, the results suggest that Chicago's TIF system does not strongly direct spending toward lower-income neighbourhoods. In raw terms, districts located in higher-income areas tend to receive more expenditure.

However, the normalized analysis provides an important clarification. Once spending is compared to the tax increment each district generates, the gap becomes much smaller, indicating that the pattern is largely driven by the financing structure of the program.

The spending difference between high- and low-income districts increased from about 3× to 6× between 2017 and 2024, suggesting that the disadvantage for lower-income communities may be intensifying.

Limitations

The income measure is an approximation: I assign midpoint values to ACS bracket counts rather than using continuous income data, which may introduce measurement error. Another limitation is that this analysis focuses on total spending at the district level, and does not examine how TIF are distributed within districts.